

Company Brochure

Daisho Chemical R&D

Innovations, Research & Engineering



DAISHO CHEMICAL R&D

Innovation, Research & Engineering

Discover Our World

Japanese Innovation. Global Expertise.

Daisho Chemical R&D is a Japanese company with over 30 years of experience in developing and supplying chemical additives and technologies for ground improvement and geotechnical engineering.

Our foundation lies in continuous research and development, which has made us not only a solutions provider but also a true partner in creating value for our clients.

Our greatest strength is the combination of an experienced technical team and a personalized, collaborative approach - offering top-quality consulting, diverse expertise, and the flexibility required to meet the demands of the industry.

DAISHO CHEMICAL R&D

Innovation, Research & Engineering

What Sets Us Apart



Practice

We have spent decades testing and refining additives specifically designed to address real-world civil engineering challenges - from deep foundations to complex urban infrastructure.

Research

We continuously develop specialized chemical admixtures and ground-improvement technologies. Our work includes the creation of the OPT Jet method and advanced formulations engineered to adapt to diverse soil conditions.



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Why Trust Daisho



Production

We deliver quality that pays off. Our high-performance chemical solutions are cost-effective, practical, and ready for large-scale use directly on the construction site.

Partnership

We believe success is achieved together. Our approach is based on close collaboration and standing side-by-side with geotechnical companies to achieve the optimal outcome for every project.



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Discover Our Products

Products:

S-Chem



For ground improvement, jet grouting, and DSM. Enhances the mixability of clay-rich, hard-to-treat soils and increases their final strength.

Deeper 25



Auxiliary agent for clay-based excavations. Prevents clay from sticking, eliminates clods, and significantly improves the strength of stabilized soil.

ACK1 - Super Retarder



Set-retarding dispersant for cement grout. Provides long-lasting fluidity, reduces bleed water, and ensures stable, high-quality ground reinforcement.

Discover Our Products

Products:

D-Retardant



A set-retarding agent for cement-soil mixtures, capable of delaying the setting time by up to 48 hours.

RAD



Supports rapid and plastic setting of cement grouts for ground anchors, performing effectively even in highly fractured ground.

D-Fluid



A high-performance superplasticizer that reduces the water content in cement grout. Improves fluidity while maintaining high strength.

Discover Our Products

Products:

Bentonite



High-quality natural clay minerals essential for constructing cast-in-place piles and slurry-supported diaphragm walls.

DK Series



Chemical solutions used in demanding conditions such as TBM tunneling, ports, marinas, and in the repair and strengthening of infrastructure.

OP Flow Series



Used in the OPT Jet method, OP-Flow improves grout fluidity, reduces slump loss, and enables uniform ground improvement.

S-Chem - we turn stiff soil into a perfectly fluid material without adding water.



Without S-CHEM

Traditional slurry: high viscosity, low efficiency, heavily coated equipment. Sticky clay drastically reduces performance and increases tool wear. Difficult cleaning directly extends cycle times and raises operating costs.

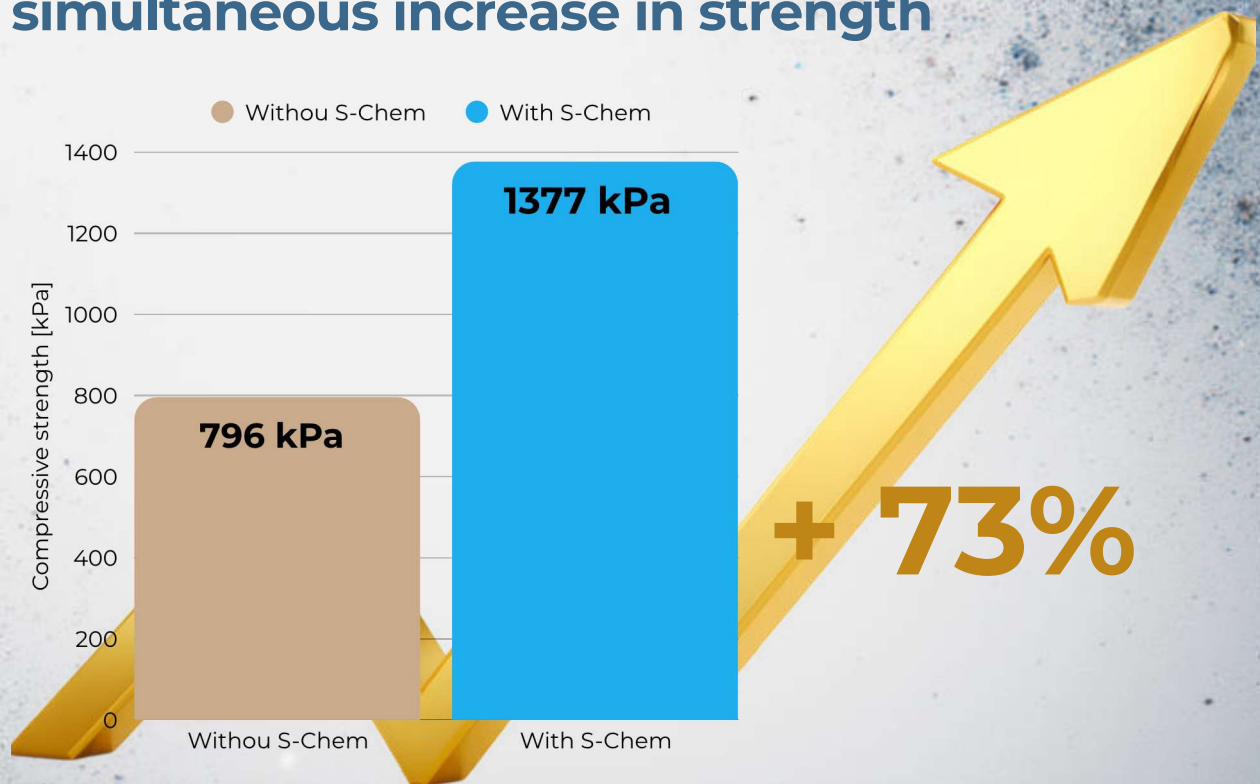


With S-CHEM

Slurry with S-Chem: optimum fluidity, significantly cleaner equipment, lower energy consumption. Reduced resistance during drilling and mixing. Lower tool wear and reduced volume of excavated spoil. Equipment stays clean, dramatically cutting downtime.

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S-Chem - improved excavability with simultaneous increase in strength



Without S-Chem

Non-uniform structure and lower final strength. The key drawback of conventional plasticizers is a significant drop in soil-cement strength.



With S-Chem

Uniform structure and higher strength at the same time. Our chemical formulations are specifically engineered to actively reinforce the soil-cement matrix, delivering higher strength at equal or reduced cement content.

DEEPER 25 - maximum shield protection



WITHOUT DEEPER 25

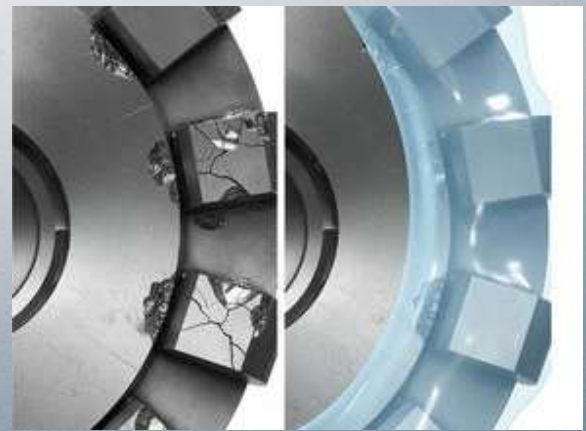


WITH DEEPER 25



Shock absorption

A special polymer forms an elastic-resilient “cushion” between the cutterhead face and the disc cutter ring. This layer effectively absorbs and dampens impact energy when striking boulders or hard rock, preventing carbide ring cracking and spalling.



Extended cutter life

By dissipating impact energy, it dramatically reduces ring cracking and uneven wear patterns. As a result, cutter change-out intervals are significantly extended, reducing downtime and overall operating costs.

DEEPER 25 - maximum shield protection



WITHOUT DEEPER 25



WITH DEEPER 25



Mixed ground

The additive simultaneously prevents clogging in clays and absorbs shocks in hard layers. It maintains stable excavation parameters even in highly variable mixed-face conditions, significantly reducing torque fluctuations and cutterhead vibration.



Extra protection

When sticky clays mix with hard boulders, unexpected cutter damage is common. DEEPER 25 instantly forms a shock-absorbing lubricating layer that buffers peak “pin-loads”, reducing metal fatigue and preventing catastrophic cutter failures.

ACK-1 Super Retarder - long-term fluidity without loss of strength



Long-term fluidity

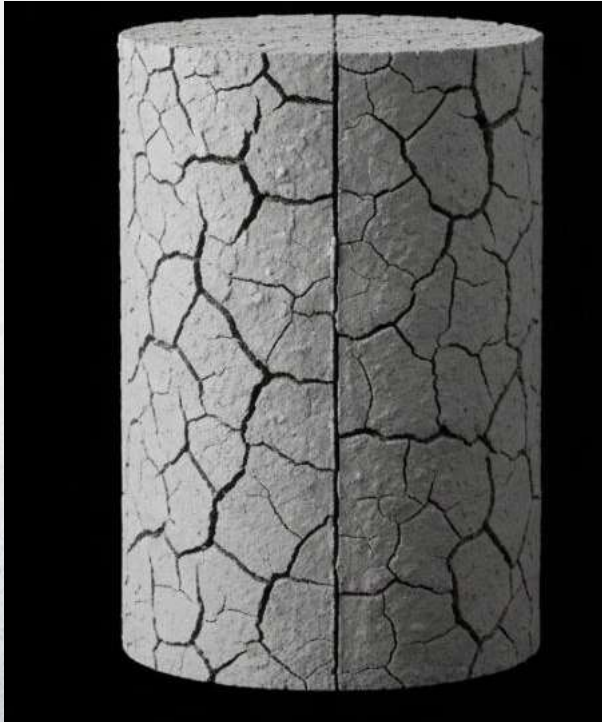
Delays cement grout setting while maintaining extremely high fluidity for many hours (even 6–8 h). This ensures uniform filling of voids during ground anchor grouting and mortar injection, with no risk of hose blockage.



Full final strength

Acts as an excellent dispersant – virtually eliminates segregation and bleeding, delivering a perfectly homogeneous hardened grout. Final compressive strength remains equal or higher compared to plain grout.

D-Retardant - setting delay of soil-cement mixture up to 48 hours



48 h full workability

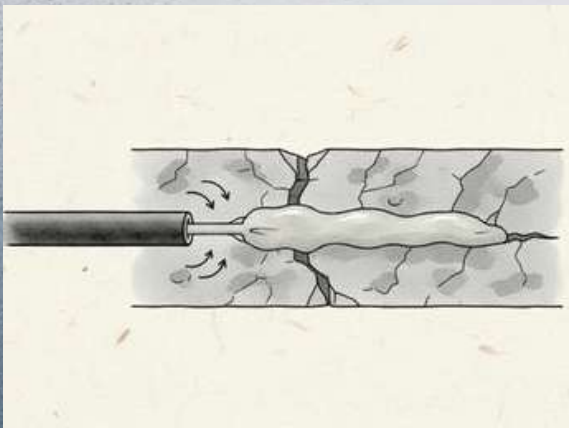
D-Retardant keeps the soil-cement mixture pumpable and fully workable for up to 48 hours - perfect for large DSM columns, railway subgrades and long tunnel sections. Enables relaxed construction scheduling without time pressure.



Full final strength

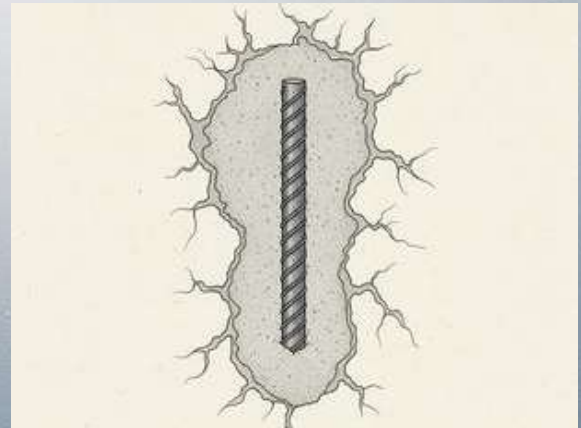
Guarantees repeatable delay while maintaining final strength equal to or higher than untreated mix. Provides perfect cement-soil blending and eliminates segregation.

RAD - rapid setting under any ground condition



Instant gelation

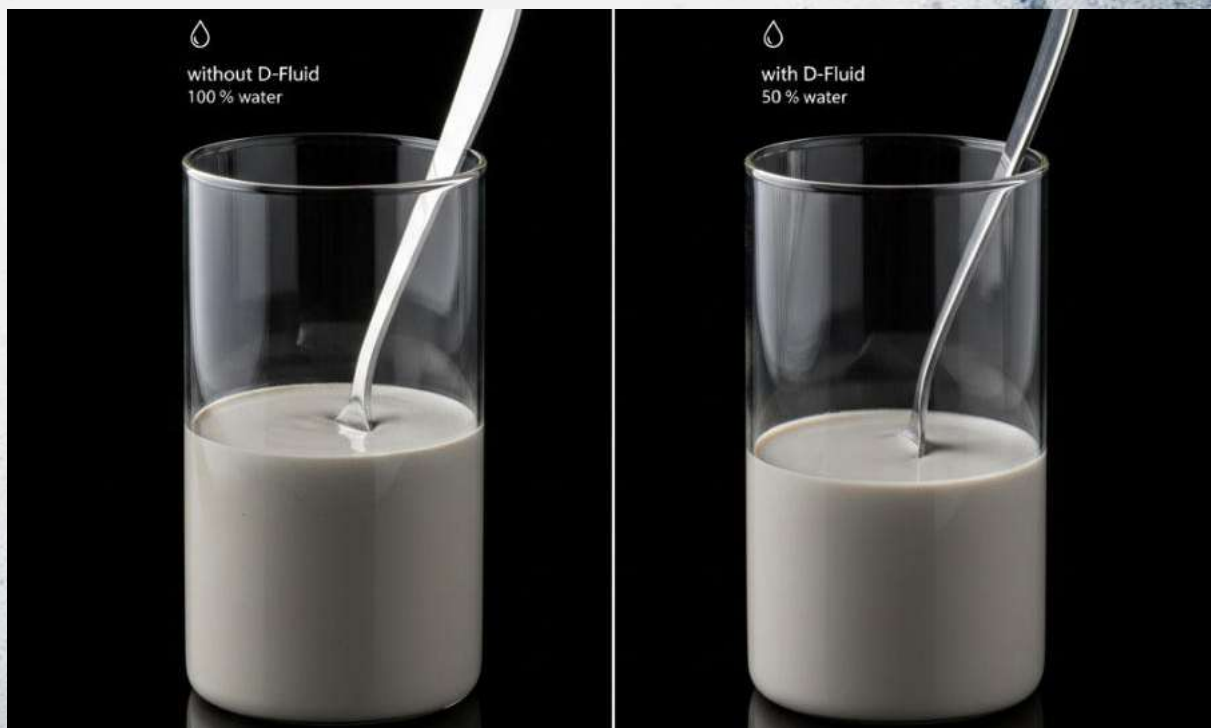
RAD causes instantaneous transition of cement grout from liquid to plastic-gel state directly inside the fracture, completely stopping outflow even in highly permeable and heavily fractured rock.



Complete void filling

Ensures 100 % filling of the annulus around the bar or rock bolt, maximum bond strength and full load-bearing capacity - no material loss and no re-grouting required.

D-Fluid - maximum fluidity with minimum water content



Superior fluidity

D-Fluid reduces water demand by up to 50 % while maintaining extreme fluidity and pumpability of cement milk for many hours.



Excellent pumpability

Grout with D-Fluid flows effortlessly through the longest hoses and smallest nozzles - perfect for ground anchors and sprayed mortar applications.

Bentonite - reliable excavation stability and safety



Reliable wall support

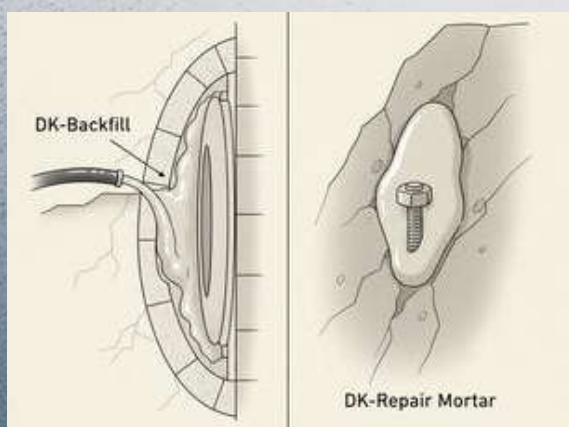
Bentonite forms a high-viscosity, thixotropic slurry that effectively supports excavation walls (diaphragm walls, secant/contiguous piles), preventing soil collapse. The trench remains fully stable throughout the entire construction phase.



Water control

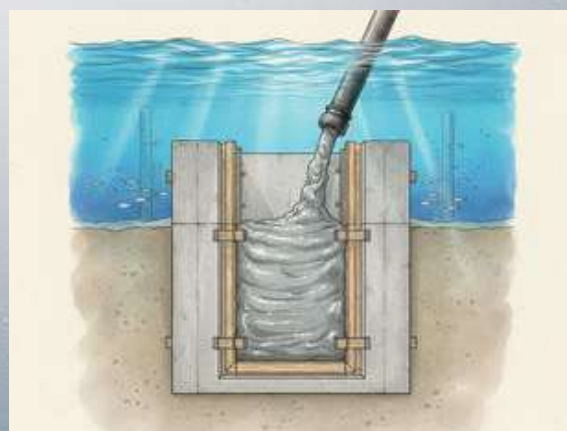
Upon hydration, bentonite swells and creates a low-permeability filter-cake on the trench walls, effectively sealing against groundwater inflow and soil fluid loss. Prevents soil loss and soil loss, protects fresh concrete from contamination and ensures long-term structural durability.

DK Series - stability on land and at sea



Backfill i Repair Mortar

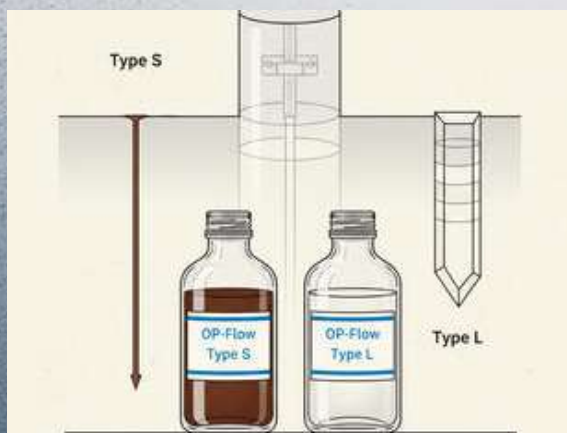
DK-Backfill - instant gelation without pipe clogging - perfect annulus and tail-void filling in TBM and shield tunnelling.
DK-Repair Mortar - high-strength repair mortar - maintains shape, no sagging or dripping even on overhead and vertical surfaces.



Aquavisco

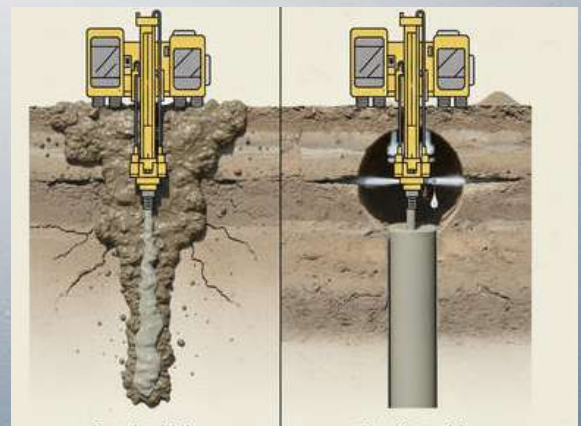
DK-Aquavisco - zero particle segregation under water - concrete retains full cohesion. No clouding and zero environmental impact - proven technology for ports, marinas and offshore foundations.

OP-Flow Series - full control in jet grouting



Type S & Type L

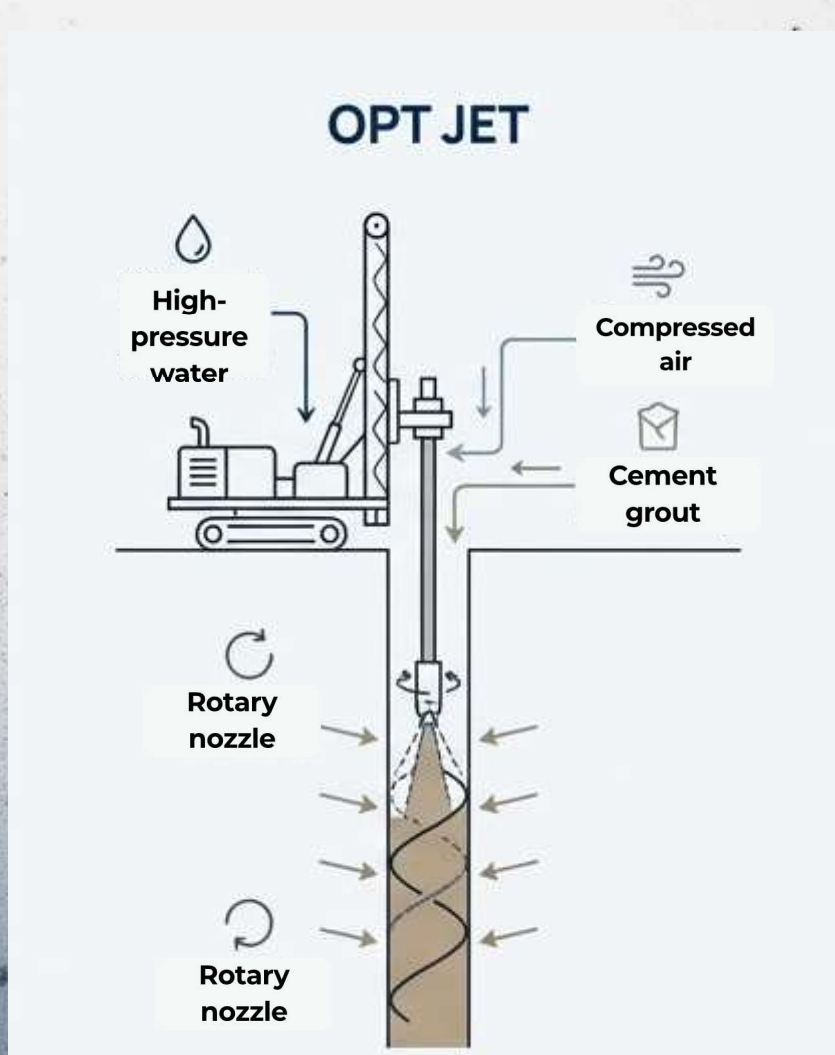
Type S - dark-brown liquid - extreme fluidity at 300 l/min and uniform soil improvement.
Type L - milky-white thixotropic liquid - instantly stops liquefaction and sudden viscosity increase in soft soils.



Innovative advantages

Dramatically reduced spoil and grout waste, sharp and highly repeatable columns even in the most challenging ground conditions. Lower cement consumption at equal or higher strength and clean job site after jetting. Developed for OPT Jet - effective in all types of jet grouting.

OPT JET - the new era of jet grouting



COST REDUCTION

- Reduced cement and water consumption
- Lower spoil volume for disposal
- Reduced equipment wear and breakdowns



TIME EFFICIENCY

- Lower resistance - faster drilling and mixing
- Extended working time thanks to controlled setting
- Shorter overall construction time and faster project delivery



SUPERIOR QUALITY

- Ability to form columns across a wide range of diameters
- Higher process predictability
- Excellent result repeatability

What is OPT Jet? An integrated system – not just a method.

OPT Jet is our proprietary jet grouting system that combines a unique double-fluid jet (ultra-high-pressure water + cement grout) with advanced chemical additives to maximise performance and quality.

Our high-performance jet grouting technology enables larger column diameters at lower lifting speeds, delivering significant time and cost savings.

With the OPT Jet system and OP-Flow chemistry we achieve larger column diameters, substantial cement savings and drastically reduced spoil volume, lowering disposal costs and improving site logistics.

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Daisho Chemical R&D -advancing geotechnical technology worldwide

Japanese heritage

"I continue the mission my father began almost 40 years ago - developing chemical additives that genuinely solve real-world geotechnical challenges. This is a heritage of quality, responsibility and innovation."

"My vision is to become a global partner in geotechnical engineering."



Sharing knowledge

"The time has come for technology developed in Japan to reach the world. Barriers are gone - today knowledge and innovation must flow freely between countries."

"We want to share our technology to help solve actual geotechnical problems - from ground stabilisation and injection works to tunnelling. Our goal is solutions that work on site, not just in the laboratory."

"We solve problems through engineering expertise and Japanese precision."



Taiju Haneda
CEO, Daisho Chemical R&D

A handwritten signature in black ink, likely belonging to Taiju Haneda.

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Next step? Let's talk!



1. Discussion

We start with a discussion about your challenges and current needs.



2. Data analysis

We begin with in-depth analysis of local soil samples to fully understand the project-specific conditions.



3. Testing & optimisation

We conduct joint laboratory tests to optimise the mix design and select the right additives to achieve target performance.



4. Implementation & support

We provide full technical support during implementation, ensuring the solution performs as designed under real site conditions.

DAISHO CHEMICAL R&D



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